

Summary and Recommendations

Data Cleaning & Preprocessing

- Imported the customer churn dataset using **Pandas** and performed an initial inspection of the data structure, data types, and missing values.
 - Replaced blank values in the **TotalCharges** column with 0 for customers having zero tenure and converted the column to a numeric data type.
 - Verified that the dataset contained **no missing values** and **no duplicate customer IDs**, ensuring data quality before analysis.
 - Converted the **SeniorCitizen** column from numeric values (0 and 1) to categorical values (**No** and **Yes**) to improve readability and visualization.
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Exploratory Data Analysis (EDA)

Overall Churn Distribution

- The dataset contains **7,043 customers**, of which **26.54% have churned** while **73.46% remain active**.
- This indicates that although the majority of customers are retained, approximately one in every four customers leaves the company.

Gender Analysis

- Customer distribution between males and females is nearly equal.
- Churn behavior is also similar across both genders, suggesting that **gender is not a significant factor influencing customer churn**.

Senior Citizen Analysis

- Senior citizens represent a smaller portion of the customer base.
- However, they exhibit a **higher proportion of churn** compared to non-senior customers, indicating they are more likely to discontinue the service.

Tenure Analysis

- Customers with **short tenure (especially within the first few months)** show the highest churn rate.

- Customers who have remained with the company for longer durations are significantly more likely to stay, highlighting the importance of early customer engagement and retention.

Contract Type Analysis

- Customers with **Month-to-Month contracts** experience the highest churn.
- Those with **One-Year** and **Two-Year contracts** demonstrate much stronger retention, indicating that longer contractual commitments reduce customer attrition.

Internet & Additional Services Analysis

- Customers using **Fiber Optic Internet** have the highest churn among all internet service categories.
- Customers who **do not subscribe to Online Security, Online Backup, Device Protection, or Tech Support** are considerably more likely to churn.
- The availability of these value-added services appears to improve customer loyalty and retention.
- **Phone Service, Multiple Lines, Streaming TV, and Streaming Movies** have comparatively little impact on churn, suggesting these services are not primary drivers of customer retention.

Payment Method Analysis

- Customers using **Electronic Check** show the highest churn rate among all payment methods.
 - Customers using **Automatic Bank Transfer** and **Automatic Credit Card** payments have much lower churn rates.
 - This suggests that customers enrolled in automatic payment methods tend to remain with the company for longer periods.
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Key Business Insights

- Approximately **one-fourth of customers have churned**, indicating a moderate customer attrition problem.
- Early-stage customers are the most vulnerable to churn, making the first few months critical for retention efforts.
- Month-to-Month contracts and Electronic Check payments are strong indicators of higher churn risk.
- Fiber Optic users require additional attention, as they experience significantly higher churn compared to DSL users.

- Customers without security and support-related services are substantially more likely to leave the company.
- Long-term contracts and automatic payment methods contribute positively to customer retention.

Recommendations

- Introduce attractive incentives to encourage customers to upgrade from Month-to-Month to long-term contracts.
- Promote **Online Security, Online Backup, Device Protection, and Tech Support** through bundled service packages.
- Investigate customer satisfaction among Fiber Optic users to identify and address potential service issues.
- Launch targeted retention campaigns for new customers during their initial months with the company.
- Encourage customers to switch from Electronic Check to automatic payment methods by offering discounts or rewards.